A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Carbonyls, carboxylic acids and chirality

The C=O group is the busiest functional group in organic chemistry. This topic covers aldehydes, ketones, carboxylic acids and the compounds made from them, the mirror-image molecules that some of their reactions produce, and the instrumental methods (NMR, mass spectrometry and chromatography) that chemists use to work out what they have made.

SPEC 15.1 Optical isomerism and the chiral centre

WHAT THIS SPEC POINT ASKS

Know that optical isomerism comes from chirality in molecules with a single chiral centre.

What makes a molecule chiral

- A **chiral centre** (asymmetric carbon) is a carbon atom bonded to **four different atoms or groups**. Mark it with an asterisk (*).
- A molecule with one chiral centre is **chiral**: it is **not superimposable on its mirror image**, like your left and right hands.
- So it exists as **two optical isomers** (enantiomers, 15.2). Optical isomerism is a type of **stereoisomerism**: same structural formula, different arrangement in space.

Finding the chiral centre

1. Ignore any carbon with two identical groups: **CH_3, CH_2** and **C=C or C=O carbons** (only three groups) can't be chiral.
2. For each remaining carbon, list the four groups. Look along each branch until the groups differ: CH_2CH_3 and $CH_2CH_2CH_3$ are different.
3. If all four are different, it's a chiral centre.

Molecule	Carbon checked	Its four groups	Chiral?
butan-2-ol, $CH_3CH(OH)CH_2CH_3$	C2	H, OH, CH_3, C_2H_5	**yes**
propan-2-ol, $CH_3CH(OH)CH_3$	C2	H, OH, CH_3, CH_3	no (two CH_3)

Molecule	Carbon checked	Its four groups	Chiral?
2-hydroxypropanoic acid (lactic acid), $CH_3CH(OH)COOH$	C2	H, OH, CH_3, COOH	**yes**
2-hydroxypropanenitrile, $CH_3CH(OH)CN$	C2	H, OH, CH_3, CN	**yes**
pentan-3-ol, $CH_3CH_2CH(OH)CH_2CH_3$	C3	H, OH, C_2H_5, C_2H_5	no (two C_2H_5)

MARK-SCHEME WORDING

- "it is non-superimposable on its mirror image"; ALLOW "there are four different atoms / groups attached to a carbon (atom)", "it has a chiral centre".
- Not chiral: "no carbon atom has 4 different groups" / "the carbon is bonded to two CH_3 / the same groups" / "it is superimposable on its mirror image".

Often asked as: explain why a compound shows optical isomerism (1–2 marks); MCQ "which compound has a chiral centre?"; explain why one product is chiral and another isn't (e.g. the HCN products of pentan-2-one and pentan-3-one).

COMMON MISTAKE

- "Four different **molecules**" attached to the carbon: it's four different **atoms or groups**.
- Listing several reasons when one is wrong: a wrong statement cancels the right one.
- Confusing a chiral centre with a **planar** molecule or a planar carbocation.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Counting several chiral centres in a bigger molecule: **menthol has 3**. With n chiral centres there can be up to **2^n** optical isomers, so menthol has 8 (WCH14 Jun 2021). The 2^n rule over-counts if the molecule is symmetrical.
- Chiral carbons "hidden" in rings, such as in the cyclic dimer of lactic acid (lactide) (WCH14 Jan 2023).

SPEC 15.2 Enantiomers and 3D diagrams

WHAT THIS SPEC POINT ASKS

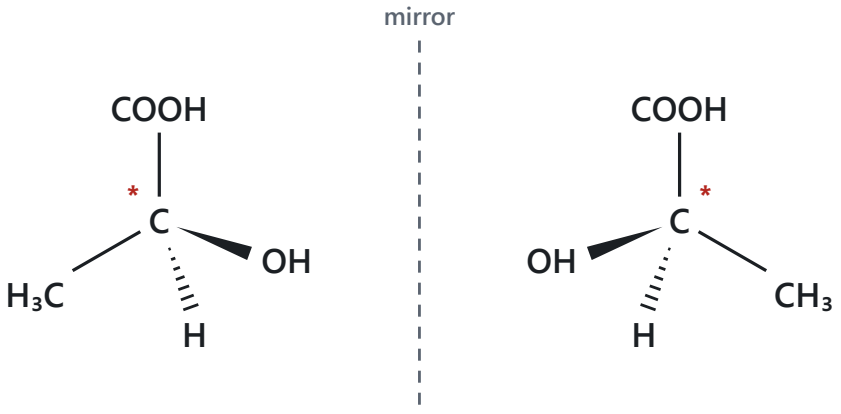
Understand that optical isomers (enantiomers) are object and non-superimposable mirror image, and draw them in 3D.

Enantiomers

- The two optical isomers are called **enantiomers**. They are **mirror images** of each other that can't be superimposed: however you turn one, it never matches the other.
- They have the **same** boiling point, melting point, density and chemical reactions with non-chiral reagents. They differ in:
  - the **direction** they rotate plane-polarised light (15.3);
  - how they react with **other chiral molecules**, such as enzymes and receptors. That's why one enantiomer of a drug can be useful and the other inactive or harmful, and why two enantiomers can smell different.

Drawing them

1. Draw the chiral carbon with a **tetrahedral** arrangement: two bonds in the plane of the paper (plain lines), one **wedge** (coming towards you) and one **hashed** bond (going away). Put the wedge and hashed bonds **next to each other**.
2. Draw a dashed **mirror line**.
3. Draw the **mirror image**: reflect every group across the line. Don't just swap two labels at random and don't draw the same isomer twice.



The enantiomers of 2-hydroxypropanoic acid (lactic acid). Each is the mirror image of the other.

MARK-SCHEME WORDING

- "a three-dimensional diagram … showing at least one dotted bond and at least one wedged bond which are next to each other", then "the mirror image of the first structure".
- Groups in any order; CO_2H for COOH is fine.

Often asked as: draw both enantiomers of lactic acid, 2-hydroxybutanenitrile or another product (2 marks).

COMMON MISTAKE

- Drawing the **same** enantiomer twice. The safest way is to draw a true reflection across a mirror line.
- Flat (2D) structures with no wedge or hashed bonds, or bonds at 90° with no 3D shape.
- Wrong atoms joined (e.g. the bond going to the H of OH): draw the bond to the atom actually attached.

SPEC 15.3 Optical activity

WHAT THIS SPEC POINT ASKS

Know that optical activity is the ability of one optical isomer to rotate the plane of polarisation of plane-polarised monochromatic light.

Plane-polarised light

- Ordinary light vibrates in all planes at right angles to its direction. A **polarising filter** lets through only light vibrating in **one plane**: **plane-polarised light**.
- **Monochromatic** light (one wavelength, e.g. from a sodium lamp) is used because the angle of rotation depends on wavelength.

Rotating the plane

- A solution of **one** enantiomer **rotates the plane** of plane-polarised light. It is **optically active**.
- The other enantiomer rotates it by the **same angle** in the **opposite direction**: one clockwise (+), one anticlockwise (−).
- A **polarimeter** measures this: light source → polariser → tube of sample solution → second polarising filter (analyser) that you rotate until the light passing through is brightest (or darkest); the angle turned is the rotation.

MARK-SCHEME WORDING

- "one isomer rotates the plane of (monochromatic) plane-polarised light in one direction and the other in the opposite direction"; ALLOW clockwise and anticlockwise.
- "Optically active" means the molecules "rotate the plane of plane-polarised light".

Often asked as: how the two enantiomers differ (1–2 marks); what "optically active" means; IP1 of a 6-mark answer on optical isomerism.

COMMON MISTAKE

- Saying the light is "reflected", "deflected" or "bent": it is the **plane of polarisation** that is **rotated**.
- Mentioning plane-polarised light but not saying the two isomers rotate it in **opposite directions**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- An SN2 product made from one enantiomer is optically active and "rotates plane-polarised light in the opposite direction" (inversion) (WCH14 Oct 2024). See 15.5.

SPEC 15.4 Racemic mixtures

WHAT THIS SPEC POINT ASKS

Know what a racemic mixture is.

Definition

- A **racemic mixture** (racemate) contains **equal amounts (equal moles) of the two enantiomers**: a 50 : 50 mixture.
- It is **optically inactive**: the rotation caused by one enantiomer is **cancelled** by the equal and opposite rotation of the other.
- A reaction that makes a chiral centre from **non-chiral** starting materials gives a racemic mixture, because there is an equal chance of forming each enantiomer (15.5). Separating them is difficult and expensive, which matters for drugs where only one enantiomer works.

MARK-SCHEME WORDING

- "(mixture containing) equal amounts of (both) enantiomers"; ACCEPT "equimolar / 50 : 50 mixture".

- Why inactive: "the rotations caused by both enantiomers cancel".
- Just stating that equal amounts of each enantiomer form scores the racemic-mixture mark.

Often asked as: define a racemic mixture; explain why a product is not optically active although it has a chiral centre.

COMMON MISTAKE

- Saying only that a racemic mixture "has no optical activity" without saying what it **contains**.
- Saying "two enantiomers" without "**equal amounts**".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Partial racemisation**: a product with more of one enantiomer than the other is optically active, but less so than the starting material. It shows the reaction went partly by SN1 and partly by SN2 (WCH14 Oct 2025).

SPEC 15.5 Optical activity as evidence for mechanisms

WHAT THIS SPEC POINT ASKS

Use the optical activity of reactants and products as evidence for SN1, SN2 and nucleophilic addition to carbonyl compounds.

Start with a single enantiomer, then look at the product

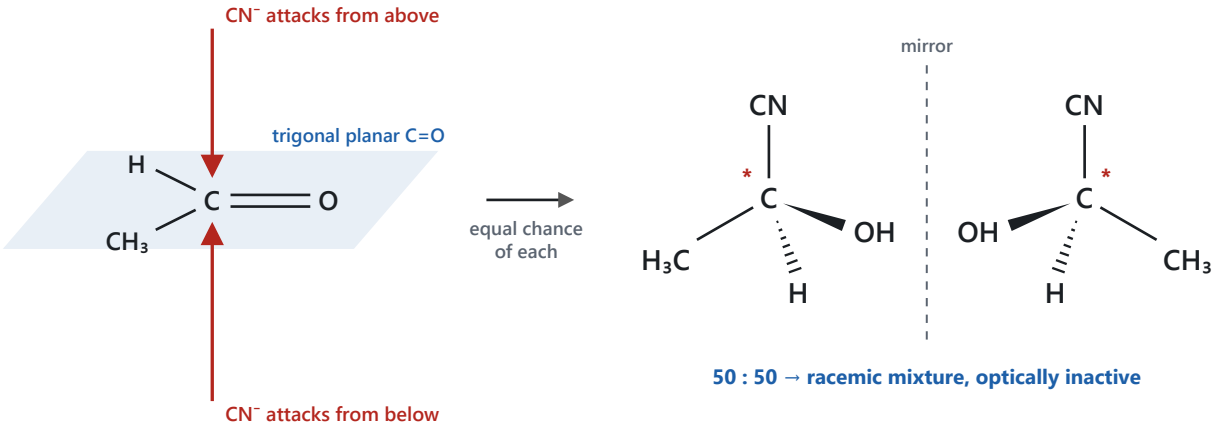
	SN1	SN2	Nucleophilic addition to C=O
Typical substrate	tertiary halogenoalkane	primary (or secondary) halogenoalkane	aldehyde or unsymmetrical ketone + HCN/CN^-
Key feature	C–X breaks first → **planar carbocation** intermediate	nucleophile attacks from the side **opposite** the halogen, in one step	the C=O group is **trigonal planar**
Attack	from **either side** of the planar carbocation, equally	only from the back	CN^- from **above or below** the plane, equally

	SN1	SN2	Nucleophilic addition to C=O
Product from one enantiomer	**racemic mixture**	**one enantiomer**, with the arrangement **inverted**	(starting material isn't chiral) **racemic mixture**
Optical activity of product	**inactive** (activity lost)	**active** (often rotates the opposite way)	**inactive**

- So if an optically active halogenoalkane gives an **inactive** (racemic) product, the mechanism was **SN1**. If the product is still **optically active**, it was **SN2**. This backs up the rate-equation evidence in 11.9.

Nucleophilic addition: why HCN gives a racemic product

- The carbonyl carbon and the three atoms attached to it are in **one plane** (trigonal planar).
- The cyanide ion can approach that plane from **above or below** with **equal probability**.
- Attack from one side gives one enantiomer; from the other side, the mirror image. So equal amounts form: a **racemic mixture**, which is **optically inactive** even though each product molecule is chiral.
- This only matters if the product has a chiral centre: **methanal** and **symmetrical ketones** (propanone, pentan-3-one) give products with two identical groups on that carbon, so no optical isomers.



CN⁻ attacks the planar C=O of ethanal equally from above and below, giving equal amounts of the two enantiomers of 2-hydroxypropanenitrile.

MARK-SCHEME WORDING

- Carbonyl: "propanal is planar around the CHO / reaction site / C=O / carbonyl"; "CN^- / nucleophile can attack on either side / both sides / above and below (giving a racemic / 50 : 50 mixture)". DO NOT AWARD "propanal is planar" alone or "planar intermediate / carbocation".

- "cyanide attacks from above and below; (as) trigonal planar around the C(δ+)". IGNORE "from both directions".
- SN2 from one enantiomer: "product is optically active; stereochemistry is inverted / product rotates plane-polarised light in the opposite direction".

Often asked as: explain why the product of HCN + an aldehyde is not optically active (2–3 marks); why one ketone gives optical isomers and another doesn't; what optical activity shows about an SN1 or SN2 reaction (6-mark or MCQ).

COMMON MISTAKE

- Writing "SN1" or "SN2" in an answer about **nucleophilic addition**: it automatically loses the marks.
- Saying "a planar **carbocation**" forms in the HCN reaction. There is no carbocation; the **C=O group** is planar.
- Saying "the molecule is planar": only the arrangement around the carbonyl carbon is.
- Giving the stock SN1 answer (racemic) for an SN2 reaction.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Which product shows optical isomerism: the HCN product of **pentan-2-one** has a chiral carbon; the product of **pentan-3-one** (two ethyl groups) doesn't. Both ketones still react (WCH14 Jan 2025).

SPEC 15.6 Naming and drawing aldehydes and ketones

WHAT THIS SPEC POINT ASKS

Name aldehydes and ketones and draw their structural, displayed and skeletal formulae.

The functional groups

- Both contain the **carbonyl group, C=O**.
- **Aldehyde**: C=O at the **end** of the chain, written **–CHO**. Name ends **-al**. The CHO carbon is always C1, so no number is needed: methanal HCHO, ethanal CH_3CHO, propanal CH_3CH_2CHO.
- **Ketone**: C=O **inside** the chain, joined to two carbons, written **–CO–**. Name ends **-one**, with a number when needed: propanone CH_3COCH_3, butanone $CH_3COCH_2CH_3$, pentan-2-one $CH_3COCH_2CH_2CH_3$, pentan-3-one $CH_3CH_2COCH_2CH_3$.
- Aldehydes and ketones with the same number of carbons are **functional group isomers**: propanal and propanone are both C_3H_6O.

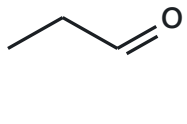
Names with substituents

- Number the chain so the C=O gets the **lowest** number, then add substituents: 2-methylpropanal $(CH_3)_2CHCHO$; 3-methylbutan-2-one $CH_3COCH(CH_3)_2$; 1-hydroxy-4-methylpentan-2-one $HOCH_2COCH_2CH(CH_3)_2$.

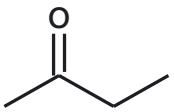
Formulae

Type	Propanal	Butanone
Molecular	C_3H_6O	C_4H_8O
Structural	CH_3CH_2CHO	$CH_3COCH_2CH_3$
Displayed	every atom and every bond drawn, including C=O and each C–H	same
Skeletal	zig-zag line ending in =O	zig-zag with =O on the second carbon

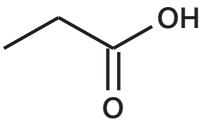
- In **skeletal** formulae every line end and corner is a carbon; H atoms on carbon are left out; O, Cl, N and the H of OH are shown.



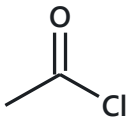
propanal
CH_3CH_2CHO



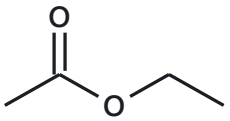
butanone
$CH_3COCH_2CH_3$



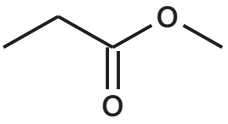
propanoic acid
CH_3CH_2COOH



ethanoyl chloride
CH_3COCl



ethyl ethanoate
$CH_3COOCH_2CH_3$



methyl propanoate
$CH_3CH_2COOCH_3$

Skeletal formulae for Topic 15. Ethyl ethanoate and methyl propanoate are isomers (both $C_4H_8O_2$): see 15.13 and 15.20.

MARK-SCHEME WORDING

- Names must have the right stem and locant. Hyphens and spaces are usually not penalised; "hydroxyl" is allowed for "hydroxy".

- Displayed formulae: "penalise missing H's only once"; structural or skeletal formulae score at most 2 when displayed was asked.
- "If more than one structure type is given, both must be correct."

Often asked as: name a carbonyl compound or its product; draw displayed formulae of products from a reaction sequence; draw skeletal formulae of isomers.

COMMON MISTAKE

- Writing propanal as CH_3CH_2COH, which looks like an alcohol: write **–CHO**.
- Ignoring the instruction "displayed" or "skeletal"; leaving out H atoms; five bonds on a carbon.
- Not writing the H of an OH group on a skeletal formula.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Hydroxynitrile names**: the nitrile carbon counts in the chain and the nitrile is the main group, so ethanal + HCN gives **2-hydroxypropanenitrile** and propanal + HCN gives **2-hydroxybutanenitrile** (WCH14 Jun 2022). Don't call them "cyanides".

SPEC 15.7 Physical properties of aldehydes and ketones

WHAT THIS SPEC POINT ASKS

Understand that aldehydes and ketones don't hydrogen bond with each other (which affects their physical properties) but can hydrogen bond with water (which affects their solubility).

Forces between carbonyl molecules

- The **C=O bond is polar** (O is more electronegative): $C^{\delta+}=O^{\delta-}$. So carbonyl molecules have **permanent dipole–dipole forces** as well as **London forces**.
- They have **no H on an O (or N or F)**, so they **cannot hydrogen bond to each other**.
- Result: boiling temperatures are **higher than alkanes** of similar Mr (extra dipole–dipole forces) but **lower than alcohols** (no hydrogen bonds).

Compound (Mr ≈ 58–60)	Strongest force between molecules	Boiling temperature / °C
butane, C_4H_{10}	London	−0.5

Compound (Mr ≈ 58–60)	Strongest force between molecules	Boiling temperature / °C
propanal, CH_3CH_2CHO	permanent dipole–dipole	49
propanone, CH_3COCH_3	permanent dipole–dipole	56
propan-1-ol, $CH_3CH_2CH_2OH$	hydrogen bonds	97
ethanoic acid, CH_3COOH	hydrogen bonds	118

Solubility in water

- The **lone pairs on the carbonyl O** can accept a hydrogen bond from the **δ+ H of a water molecule**. So small aldehydes and ketones dissolve well (propanone mixes with water in all proportions).
- As the carbon chain gets longer, solubility **falls**: the non-polar chain can't hydrogen bond with water, and London forces between the bigger molecules get stronger.

MARK-SCHEME WORDING

- "pentan-2-one does not form hydrogen bonds (between molecules)"; "hydrogen bonds require more energy to break (so pentan-2-one has a lower boiling temperature)". ALLOW "(only) permanent dipole–dipole and London forces". IGNORE "pentan-2-ol has stronger London forces". DO NOT AWARD "more hydrogen bonds".
- "propanal forms permanent dipole–permanent dipole forces" (not hydrogen bonds).

Often asked as: compare the boiling temperatures of a carbonyl and an alcohol of similar Mr (2 marks); 6-mark comparing an aldehyde, an alcohol and a carboxylic acid; MCQ on forces.

COMMON MISTAKE

- Saying an aldehyde or ketone hydrogen bonds **with itself**.
- Naming the forces in each compound without **comparing their strengths** and linking to the **energy needed**.
- Talking about hydrogen bonding with **water** when the question is about boiling temperature.
- Saying covalent bonds break on boiling.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- A hydrogen bond needs an H on N/O/F **on one side only**: an ester or ketone O can accept one from water even though the molecule has no O–H (WCH14 Jan 2022).

- London forces are similar when molecules have a similar number of electrons, so you can compare the other forces directly (WCH14 Jan 2021).

SPEC 15.8 Reactions of carbonyl compounds

WHAT THIS SPEC POINT ASKS

Know the reactions of aldehydes and ketones with: (i) Fehling's/Benedict's, Tollens' and acidified dichromate(VI); (ii) $LiAlH_4$ in dry ether; (iii) HCN with KCN (mechanism, including optical activity); (iv) 2,4-DNPH, including identification from derivative melting temperatures; (v) iodine in alkali (iodoform test).

(i) Telling aldehydes from ketones: oxidation

- **Aldehydes** have an H on the carbonyl carbon, so they are **easily oxidised** to carboxylic acids: RCHO + [O] → RCOOH.
- **Ketones** have no such H and are **not oxidised** by these mild reagents (it would need a C–C bond to break).
- The aldehyde **reduces** the reagent; the colour change shows it.

Reagent and conditions	Aldehyde	Ketone	What is reduced
Fehling's or **Benedict's** solution, warm	blue solution → **brick-red precipitate**	stays blue	$Cu^{2+} \rightarrow Cu_2O$ (copper(I) oxide)
Tollens' reagent (ammoniacal silver nitrate), warm in a water bath	**silver mirror** (or black/grey precipitate)	no change	$Ag^+ \rightarrow Ag$: $[Ag(NH_3)_2]^+ + e^- \rightarrow Ag + 2NH_3$
Acidified potassium dichromate(VI) ($K_2Cr_2O_7$ + dilute H_2SO_4), warm	**orange → green**	stays orange	$Cr_2O_7^{2-} \rightarrow Cr^{3+}$

- Equation with [O]: $CH_3CH_2CHO + [O] \rightarrow CH_3CH_2COOH$. (Fehling's and Tollens' are alkaline, so the carboxylate ion actually forms.)
- Acidified dichromate also oxidises primary and secondary **alcohols**, so it is not a test for aldehydes if alcohols might be present. Fehling's and Tollens' don't react with alcohols.

MARK-SCHEME WORDING

- Test + positive result for the aldehyde + "no change" for the ketone; M2 depends on M1. ALLOW "ammoniacal silver nitrate", "$Cr_2O_7^{2-}$ and H^+". DO NOT AWARD hydrochloric acid.
- Observations: "brick-red / red / brownish red precipitate (from blue solution)"; "silver mirror" (ALLOW black/grey precipitate); "goes green / blue (from orange)".

- "No change", "no result" and "no observation" are equivalent.

Often asked as: one test to distinguish an aldehyde from a ketone (3 marks); part of a multi-test identification (6-mark); equation using [O].

COMMON MISTAKE

- Mixing up Tollens' (silver mirror) and Fehling's/Benedict's (red precipitate).
- Leaving out "**acidified**" before dichromate.
- Adding water as a product in RCHO + [O] → RCOOH.
- Writing just "positive" or "it reacts": give the **observation**.

(ii) Reduction with $LiAlH_4$

- **Reagent**: lithium tetrahydridoaluminate(III), **$LiAlH_4$**, in **dry ether** (ethoxyethane), at room temperature. Then add water or dilute acid to finish. Water must be absent during the reduction because $LiAlH_4$ reacts violently with it.
- Use **[H]** in equations:
  - **aldehyde → primary alcohol**: $CH_3CH_2CHO + 2[H] \rightarrow CH_3CH_2CH_2OH$ (propan-1-ol)
  - **ketone → secondary alcohol**: $CH_3COCH_3 + 2[H] \rightarrow CH_3CH(OH)CH_3$ (propan-2-ol)
- $LiAlH_4$ supplies **hydride ions, H^-**, which act as nucleophiles: H^- attacks the δ+ carbonyl carbon and the O^- formed is then protonated. It attacks C=O but not C=C, because C=C is **non-polar** (no δ+ carbon to attract H^-).

MARK-SCHEME WORDING

- "lithium tetrahydridoaluminate / lithium aluminium hydride / $LiAlH_4$"; "dry and ether / ethoxyethane". ALLOW "anhydrous ether". DO NOT AWARD heat or reflux. Name the product with its number (propan-1-ol, not "propanol").
- Why C=O and not C=C: "the C=O bond is polarised by the (electronegative) oxygen"; "H^- is attracted to the δ+ carbon"; "the C=C bond is non-polar".

Often asked as: reagent and conditions (2 marks); name the products from an aldehyde and a ketone; balanced equation with [H]; MCQ picking the reducing agent.

COMMON MISTAKE

- "Dry ester" for dry ether, or adding heat/reflux.
- Calling the reaction "nucleophilic addition" when asked for the type: the spec classes it as **reduction**.
- Unbalanced equations: two [H] per C=O reduced.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

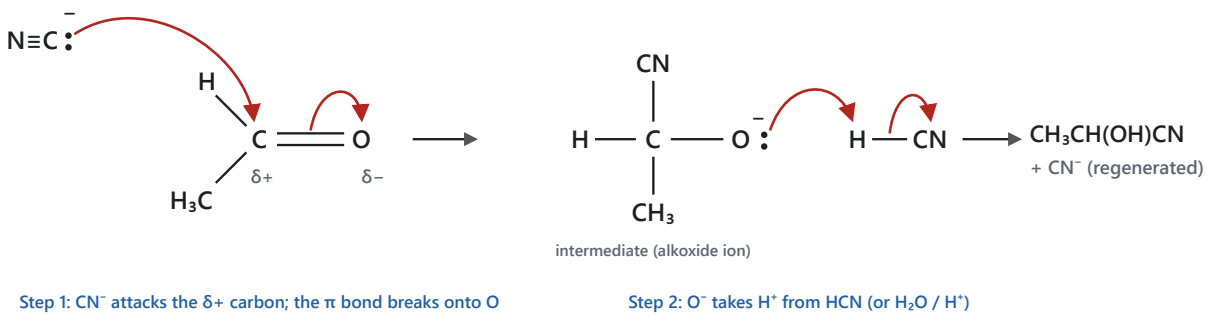
- The **mechanism** of hydride addition: dipole on C=O, lone pair on H^-, arrow from H^- to the carbonyl C, arrow from the C=O bond to O; then the O^- is protonated by acid (WCH14 Jan 2023).

(iii) Nucleophilic addition of HCN

- **Reagents**: **HCN** in the presence of **KCN** (or KCN with a little dilute acid), at room temperature. Both are **highly toxic**: use a fume cupboard.
- **Why KCN?** HCN is a very weak acid, so it hardly dissociates and gives very little CN^-. KCN supplies plenty of **CN^-**, which is the real **nucleophile**. CN^- is regenerated in step 2, so it acts as a **homogeneous catalyst**.
- **Product**: a **2-hydroxynitrile** (hydroxynitrile). The carbon chain **grows by one carbon**, which is useful in synthesis.
  - ethanal + HCN → **2-hydroxypropanenitrile**, $CH_3CH(OH)CN$
  - propanone + HCN → **2-hydroxy-2-methylpropanenitrile**, $(CH_3)_2C(OH)CN$

The mechanism, step by step

1. The C=O bond is **polar**: show **δ+ on C** and **δ− on O**.
2. **Step 1**: a **lone pair on the carbon of CN^-** attacks the δ+ carbonyl carbon (curly arrow from the lone pair to the C). At the same time the **π bond of C=O breaks**, with both electrons moving to the O (curly arrow from the C=O bond to the O).
3. This gives a negatively charged **intermediate** (an alkoxide ion, with O^-).
4. **Step 2**: a **lone pair on O^-** attacks the **H of an HCN molecule** (or H^+ / H_2O); the H–C bond in HCN breaks, re-forming CN^-.
5. Product: the 2-hydroxynitrile.



Nucleophilic addition of HCN to ethanal. Four curly arrows: CN^- lone pair → C; C=O bond → O; O^- lone pair → H of HCN; H–CN bond → C of CN.

- Why "nucleophilic addition": the attacking species is a **nucleophile** (CN^-, an electron-pair donor attracted to the δ+ carbon), and two species **add** across the C=O to give **one** product, with nothing removed.
- With an aldehyde (other than methanal) or an unsymmetrical ketone the product is a **racemic mixture**: see 15.5.

MARK-SCHEME WORDING

- Mechanism marks: dipole on C=O; arrow from the C=O bond to O; **lone pair on the C of CN^-** and the arrow from it to the carbonyl carbon; intermediate with its negative charge; lone pair on O^- and arrow to the H of HCN (or H^+); product.
- Role of KCN: "catalyst"; "homogeneous" = same phase as the reactants.
- Role of NaOH (if used instead of KCN): "reacts with HCN to produce CN^- / moves the dissociation equilibrium of HCN to the right / increases $[CN^-]$". IGNORE "speeds up the reaction".
- "Justify nucleophilic addition": "the (negative) cyanide ion attacks a δ+ centre / region of low electron density"; "nothing is lost / only one product".

Often asked as: the full mechanism (4 marks, very common); reagents for adding one carbon; role of KCN or NaOH; the rate equation; why the product is not optically active.

COMMON MISTAKE

- Arrows that start in empty space or end on a bond instead of an atom.
- No lone pair on CN^-, or the lone pair drawn on **N** with the bond made from C.
- Missing negative charges on CN^- and on the O of the intermediate; a **positive** charge on the carbonyl C (it's only δ+).
- Using the wrong carbonyl compound (read the stem carefully).
- Explaining "nucleophilic addition" by just repeating the words "nucleophile" and "add".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Rate equation**: rate = $k[CH_3CHO][CN^-]$. HCN is **not** in it, because step 1 (CN^- attack) is the slow step (WCH14 Oct 2024).
- A **diketone** adds HCN at **both** C=O groups: $CH_3C(OH)(CN)CH_2CH_2C(OH)(CN)CH_3$ from hexane-2,5-dione (WCH14 Jun 2023).

(iv) 2,4-Dinitrophenylhydrazine (2,4-DNPH, Brady's reagent)

- **Test for a carbonyl group (C=O in aldehydes and ketones)**: add 2,4-DNPH solution. An **orange (or yellow/red) precipitate** forms. Both aldehydes and ketones give it.
- Carboxylic acids and esters also contain C=O but **don't** react.

- The reaction is a **condensation** (addition, then loss of water). You don't need the equation.

Identifying the carbonyl compound from the derivative

1. **Filter** off the orange precipitate (Buchner funnel, reduced pressure).
2. **Recrystallise** it to purify:
   - dissolve in the **minimum volume of hot solvent** (so the solution is saturated and as much as possible crystallises on cooling);
   - **cool** (then in ice) so crystals of the derivative form while soluble impurities stay in solution;
   - **filter** and **wash with a little ice-cold solvent** (removes the impure solution without dissolving many crystals);
   - **dry** (between filter papers, in a low oven or a desiccator).
3. Measure the **melting temperature** and compare it with a **data table** of known 2,4-DNPH derivatives. A pure compound melts **sharply** at a value matching the table.

WORKED EXAMPLE

Worked example: identifying a carbonyl from its derivative

An unknown C_4H_8O gives an orange precipitate with 2,4-DNPH and no change with Tollens' reagent. After recrystallisation the derivative melts at 115–116 °C.

Melting temperatures of 2,4-DNPH derivatives (example data): butanal 123 °C; butanone 116 °C; 2-methylpropanal 182 °C.

The negative Tollens' test shows a **ketone**, and the melting temperature matches **butanone**.

MARK-SCHEME WORDING

- "2,4-dinitrophenylhydrazine / 2,4-DNPH" (ALLOW Brady's reagent); "orange precipitate" (ALLOW yellow / red; DO NOT AWARD reddish-brown).
- "the solids need to be recrystallised"; "their melting temperatures determined and compared with data / reference books". DO NOT AWARD **boiling** temperature. IGNORE the colour of the precipitate as a way of identifying.
- Full outline: filter; dissolve in minimum hot solvent and cool; rinse with ice-cold solvent; dry; melting temperature compared with data.

Often asked as: test for a carbonyl; how to identify which carbonyl it is (2–3 marks, or a 6-mark on recrystallisation); link a positive 2,4-DNPH and a negative Tollens' to a ketone.

COMMON MISTAKE

- **Boiling** temperature instead of **melting** temperature: very common.
- "Purify it" without saying **recrystallise**; "fractional distillation" for a solid.
- Using the colour of the precipitate to tell carbonyls apart.

- Thinking 2,4-DNPH reacts only with aldehydes.
- Suggesting NMR or IR when the question asks for a school-lab chemical method.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The **purification and identification routine**: filter, recrystallise, measure the melting temperature, never the boiling temperature, then compare with data. This has been asked on four papers (WCH14 Oct 2024).

(v) The iodoform (triiodomethane) test

- **Reagents**: **iodine** solution and **sodium hydroxide** solution (alkaline iodine), warm gently.
- **Positive result**: a **pale yellow precipitate** of **triiodomethane, CHI_3** (iodoform), which has an **antiseptic smell**.
- **Positive for**: compounds with a **$CH_3CO–$** group (**methyl ketones** and **ethanal**, the only aldehyde).
- Also positive for alcohols with a **$CH_3CH(OH)–$** group (ethanol and secondary alcohols such as propan-2-ol and butan-2-ol), because the alkaline iodine first oxidises them to $CH_3CO–$ compounds.
- **Negative for**: methanal and other aldehydes; ketones without a methyl next to the C=O (e.g. pentan-3-one); methanol; primary alcohols other than ethanol; tertiary alcohols.

Equation (propanone): $CH_3COCH_3 + 3I_2 + 4NaOH \rightarrow CH_3COONa + CHI_3 + 3NaI + 3H_2O$

- The CH_3 group becomes CHI_3; the rest of the molecule becomes the **sodium salt of a carboxylic acid with one fewer carbon**.

Compound	Iodoform test
ethanal, CH_3CHO	positive
propanal, CH_3CH_2CHO	negative
propanone, CH_3COCH_3	positive
pentan-2-one, $CH_3COCH_2CH_2CH_3$	positive
pentan-3-one, $CH_3CH_2COCH_2CH_3$	negative
ethanol, CH_3CH_2OH	positive
propan-2-ol, $CH_3CH(OH)CH_3$	positive
propan-1-ol, $CH_3CH_2CH_2OH$	negative

MARK-SCHEME WORDING

- "iodine / I_2 and alkali / NaOH / KOH"; "(pale) yellow precipitate / crystals" for the methyl ketone and no change for the other; ALLOW "iodoform / CHI_3 / triiodomethane" and "antiseptic smell". DO NOT AWARD "yellow-orange".
- Equation marks: CHI_3; CH_3COONa; the rest balanced with state symbols.
- "S contains a methyl secondary alcohol group … as it reacts with iodine in sodium hydroxide".

Often asked as: distinguish pentan-2-one from pentan-3-one (2 marks); observation; balanced equation; which alcohols give a positive result (MCQ).

COMMON MISTAKE

- "Yellow solution" or "turns yellow": iodine is already yellow-brown. You need a **precipitate**.
- Giving **CHI_3** as the group that is present: the group is **$CH_3CO–$** (or $CH_3CH(OH)–$).
- Two observations when one is wrong (e.g. "yellow precipitate, blue-black colour"): the wrong one cancels the mark.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The test is positive for **$CH_3CH(OH)–$** alcohols too, which lets you tell propan-2-ol from propan-1-ol (WCH14 Oct 2024).
- **Methanal** is an aldehyde, so it gives a positive Fehling's/Benedict's test, but it has no CH_3CO group, so no iodoform. Ethanal is the only aldehyde that gives both (WCH14 Oct 2022).

SPEC 15.9 Naming and drawing carboxylic acids

WHAT THIS SPEC POINT ASKS

Name carboxylic acids and draw their structural, displayed and skeletal formulae.

Naming

- The **carboxyl group, –COOH**, is always at the **end** of a chain. Its carbon is **C1** and is counted in the stem.
- Name ends **-oic acid**: methanoic acid HCOOH, ethanoic acid CH_3COOH, propanoic acid CH_3CH_2COOH, butanoic acid $CH_3CH_2CH_2COOH$. General formula **$C_nH_{2n+1}COOH$**.
- Substituents are numbered from the COOH carbon: 2-hydroxypropanoic acid (lactic acid) $CH_3CH(OH)COOH$; 2-methylpropanoic acid $(CH_3)_2CHCOOH$; 3-hydroxy-2-methylbutanoic acid $CH_3CH(OH)CH(CH_3)COOH$.

- Two COOH groups: **-dioic acid**, e.g. ethanedioic acid HOOC–COOH; benzene-1,4-dicarboxylic acid HOOC–C_6H_4–COOH.
- Unsaturated acids: the COOH carbon is still C1, so CH_2=$CHCH_2COOH$ is **but-3-enoic acid**.

Formulae

- Structural: CH_3CH_2COOH (or $CH_3CH_2CO_2H$).
- Displayed: show C=O and the O–H bond separately.
- Skeletal: see the figure in 15.6; the =O and OH are drawn on the last carbon.

MARK-SCHEME WORDING

- "but-3-enoic acid"; "2-methylpropenoic acid"; IGNORE missing hyphens, spaces, commas, capitals; ACCEPT "3-butenoic acid".
- "2-hydroxypropanoic acid"; ALLOW "propaneoic"; DO NOT AWARD "hydroxo".

Often asked as: name an acid from its structure; draw or name isomers of a C_4 acid.

COMMON MISTAKE

- Numbering the alkene as position 1: the **COOH carbon** must get the lowest number.
- Forgetting that the COOH carbon is part of the stem (CH_3CH_2COOH is propanoic, not ethanoic, acid).
- Drawing carbons with five bonds when listing isomers.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Isomers of unsaturated acids, including **E/Z** isomers and a ring isomer such as cyclopropanecarboxylic acid (WCH14 Jun 2024).

SPEC 15.10 Hydrogen bonding in carboxylic acids

WHAT THIS SPEC POINT ASKS

Understand how hydrogen bonding affects the boiling temperatures and solubility of carboxylic acids.

Boiling temperatures

- The –COOH group has an **O–H** bond (δ+ H) and two O atoms with lone pairs, so carboxylic acid molecules form **hydrogen bonds** with each other, as well as London and dipole–dipole forces.
- So their boiling temperatures are **high**: ethanoic acid (Mr 60) boils at 118 °C, well above propanal (49 °C) and propan-1-ol (97 °C) of similar Mr.
- More energy is needed to overcome the hydrogen bonds between the molecules.

Solubility

- The –COOH group hydrogen bonds with **water** (both through its O–H and through the lone pairs on its O atoms), so the small acids (C_1–C_4) **mix completely** with water.
- As the **non-polar hydrocarbon chain** gets longer, solubility **falls**: the chain can't hydrogen bond with water, and London forces between the large molecules are stronger.

Drawing a hydrogen bond to water

- Show the **δ+ H** of one molecule lined up with a **lone pair on an O** of the other, the H···O bond **straight (about 180°)**, labelled "hydrogen bond", with **δ+ and δ−** on the atoms involved and on the atoms bonded to them.

MARK-SCHEME WORDING

- Hydrogen bond diagram (3 marks for five points): water drawn correctly and aligned at "160–200°"; "hydrogen bond" labelled; "lone pair on the relevant O lined up with the hydrogen bond"; δ+ and δ− on the two atoms in the hydrogen bond; and on the atoms bonded to them.
- Solubility: "can form hydrogen bonds (with water)"; "the larger the (non-polar hydrocarbon) chain the lower the solubility" because "London forces between larger acid molecules are stronger (more electrons)".
- Boiling: the acid "forms hydrogen bonds but the others do not / hydrogen bonds are the strongest intermolecular force".

Often asked as: draw a hydrogen bond between an acid and water; explain the trend in solubility; rank the boiling temperatures of an acid, an alcohol and an aldehyde.

COMMON MISTAKE

- Hydrogen bond drawn to the H of water from the H of the acid, or at an angle; no lone pair shown.
- Writing water as HO_2.
- Explaining boiling temperature with hydrogen bonding to water.

EXAM TIP

You may meet the idea that carboxylic acids form **dimers** (pairs held by two hydrogen bonds), which helps explain why their boiling temperatures are even higher than alcohols of similar Mr. It hasn't been examined: the marks are for "hydrogen bonds between molecules need more energy to break".

SPEC 15.11 Preparing carboxylic acids

WHAT THIS SPEC POINT ASKS

Know that carboxylic acids can be made by oxidising primary alcohols or aldehydes, and by hydrolysing nitriles.

Oxidation of primary alcohols and aldehydes

- **Reagents and conditions**: **potassium dichromate(VI)** and **dilute sulfuric acid** (acidified $K_2Cr_2O_7$), in **excess**, **heat under reflux**. Colour change **orange** → **green**.
- Reflux lets the reaction run hot for a long time without losing volatile material, so the aldehyde formed first is oxidised all the way to the acid. (If you **distil** the mixture as it forms instead, you collect the **aldehyde**.)
- Equations:
  - $CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O$ (primary alcohol → acid)
  - $CH_3CHO + [O] \rightarrow CH_3COOH$ (aldehyde → acid)
- Secondary alcohols give **ketones** (no further oxidation); tertiary alcohols are **not oxidised**.

Hydrolysis of nitriles

- **Acid hydrolysis**: heat under reflux with **dilute hydrochloric (or sulfuric) acid**:
  $CH_3CH_2CN + 2H_2O + HCl \rightarrow CH_3CH_2COOH + NH_4Cl$
  (ionic version: $RCN + 2H_2O + H^+ \rightarrow RCOOH + NH_4^+$)
- **Alkaline hydrolysis**: reflux with NaOH(aq) gives the **salt** and ammonia: $CH_3CH_2CN + H_2O + NaOH \rightarrow CH_3CH_2COONa + NH_3$. Then add a strong acid to release the free carboxylic acid.
- Combined with 15.8(iii), this gives a two-step route from a carbonyl to a **2-hydroxy acid** with one extra carbon: CH_3CHO → (HCN, KCN) → $CH_3CH(OH)CN$ → (dilute HCl, reflux) → $CH_3CH(OH)COOH$ (lactic acid)

MARK-SCHEME WORDING

- "potassium dichromate((VI)) / $K_2Cr_2O_7$ and sulfuric acid / H_2SO_4; (heat under) reflux". ALLOW $Cr_2O_7^{2-}$ and H^+, sodium dichromate. DO NOT AWARD hydrochloric acid or another reagent.
- Nitrile: "hydrolysis; (dilute) hydrochloric acid / HCl(aq); heat (under reflux)"; equation $(CH_3)_2C(OH)CH_2CN + 2H_2O + HCl \rightarrow (CH_3)_2C(OH)CH_2COOH + NH_4Cl$. Leaving out water or HCl, or giving HCN as a product, loses the mark.

Often asked as: reagents and conditions for oxidation; equation for nitrile hydrolysis; a two-step synthesis via a hydroxynitrile.

COMMON MISTAKE

- Calling the sulfuric acid a **catalyst** (it's a reagent here); using HCl or $KMnO_4$ with dichromate.
- Distillation instead of reflux when the acid is wanted.
- Nitrile hydrolysis products such as HCN, or no NH_4Cl.
- For the second step of a synthesis, giving an oxidising agent instead of dilute acid.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Two-step synthesis**: carbonyl → hydroxynitrile → hydroxy acid, with reagents for each step (WCH14 Jan 2024).

SPEC 15.12 Reactions of carboxylic acids

WHAT THIS SPEC POINT ASKS

Know the reactions of carboxylic acids with $LiAlH_4$ in dry ether, with bases to make salts, with PCl_5, and with alcohols (acid catalyst).

(i) Reduction with $LiAlH_4$

- **$LiAlH_4$ in dry ether**, room temperature, then water/dilute acid. The acid is reduced to a **primary alcohol**: $CH_3COOH + 4[H] \rightarrow CH_3CH_2OH + H_2O$
- Four [H] are needed (C=O and C–OH both changed), and water is formed.

(ii) With bases: salts

- Carboxylic acids are **weak acids**, but strong enough to react with:
  - **alkalis**: $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$
  - **carbonates and hydrogencarbonates**, with **effervescence** of CO_2 (turns limewater milky):
    $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$
    $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + CO_2 + H_2O$
  - **reactive metals**: $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$
- Fizzing with Na_2CO_3 or $NaHCO_3$ is a **test for –COOH**: alcohols and phenols don't do it.

- The salts are **ionic**: write $CH_3COO^-Na^+$ or CH_3COONa, never a covalent O–Na bond. Name: sodium ethanoate.

(iii) With phosphorus(V) chloride, PCl_5

- Add solid **PCl_5** to the dry acid at room temperature: **steamy (misty) fumes of HCl**. The product is an **acyl chloride**:
  $CH_3COOH + PCl_5 \rightarrow CH_3COCl + POCl_3 + HCl$
- PCl_5 reacts with **any –OH** group, so alcohols give the same steamy fumes ($ROH + PCl_5 \rightarrow RCl + POCl_3 + HCl$). It is a test for OH, not specifically for COOH.

(iv) With alcohols: esterification

- Heat the carboxylic acid and alcohol under **reflux** with a few drops of **concentrated sulfuric acid** as a **catalyst**:
  $CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOCH_2CH_3 + H_2O$ (ethyl ethanoate)
- It is a **condensation** (water lost) and is **reversible**, so the yield is limited. To push the equilibrium right: use an **excess** of one reactant, or remove the ester or water as it forms. Conc. H_2SO_4 also absorbs some water.
- Esters have **sweet, fruity smells**: used in flavourings, perfumes and as solvents.

CORE PRACTICAL

Making and purifying an ester (suggested practical)

1. Mix ethanol and ethanoic acid in a round-bottomed flask, add a few drops of concentrated H_2SO_4 and anti-bumping granules.
2. **Heat under reflux** for about 20 minutes.
3. **Distil** off the crude ester (with some ethanol, acid and water).
4. Shake the distillate with **sodium carbonate solution** in a **separating funnel** to remove acids (release the pressure: CO_2 forms). Run off and discard the lower aqueous layer.
5. Shake with **calcium chloride solution** to remove ethanol; discard the aqueous layer.
6. **Dry** the ester layer with **anhydrous calcium chloride** or **anhydrous magnesium sulfate** until clear.
7. **Redistil**, collecting the fraction that boils near the ester's boiling temperature (ethyl ethanoate 77 °C).

MARK-SCHEME WORDING

- $LiAlH_4$: "dry and ether"; product a primary alcohol.
- Salt: $CH_3CH_2CH_2COOH + NaHCO_3 \rightarrow CH_3CH_2CH_2COONa + CO_2 + H_2O$; ALLOW the ionic form and H_2CO_3 for $CO_2 + H_2O$. DO NOT AWARD a covalent O–Na bond.
- PCl_5: "PCl_5 / phosphorus(V) chloride / phosphorus pentachloride"; "steamy / misty fumes"; DO NOT AWARD PCl_5(aq) or "fruity smell".
- Esterification: "(concentrated) sulfuric acid"; its role is "catalyst" (DO NOT AWARD just "proton donor").

Often asked as: equations for each reaction; test results that identify a carboxylic acid (fizzing with carbonate); reagents to make a named ester; MCQ choosing the reducing agent; identify the acid and alcohol that make a given ester.

COMMON MISTAKE

- O–Na drawn as a covalent bond in a sodium carboxylate.
- "Sodium" or "potassium" as the metal that gives fizzing in a test: use magnesium or a carbonate.
- Forgetting that PCl_5 also reacts with an alcohol OH elsewhere in the molecule.
- Unbalanced PCl_5 equations (a diacid needs $2PCl_5$).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- A **hydroxy acid** reacts with $NaHCO_3$ at the COOH only; the alcohol OH doesn't react: $CH_3CH(OH)COOH + NaHCO_3 \rightarrow CH_3CH(OH)COO^-Na^+ + H_2O + CO_2$ (WCH14 Jun 2021).
- With **excess PCl_5**, both the COOH and an alcohol OH in the same molecule are converted to –Cl (WCH14 Jun 2024). A **dicarboxylic acid** gives a diacyl chloride: $HOOCCOOH + 2PCl_5 \rightarrow ClOCCOCl + 2POCl_3 + 2HCl$ (WCH14 Jan 2025).

SPEC 15.13 Naming acyl chlorides and esters

WHAT THIS SPEC POINT ASKS

Name acyl chlorides and esters and draw their structural, displayed and skeletal formulae.

Acyl chlorides

- The –OH of the carboxylic acid is replaced by –Cl: functional group **–COCl**.
- Name: change **-oic acid** to **-oyl chloride**. Ethanoyl chloride CH_3COCl; propanoyl chloride CH_3CH_2COCl; benzoyl chloride C_6H_5COCl.

Esters

- Functional group **–COO–** (written RCOOR′).
- Name = **alkyl** group from the **alcohol** + **-oate** from the **acid**: "alkyl alkanoate".
- To name an ester from its formula, split it at the **single-bonded O**:
  - the part **with the C=O** comes from the acid → "…anoate";

- the part joined to the other O comes from the alcohol → "…yl", written **first**.

Ester	Acid part	Alcohol part	Name
$CH_3COOCH_2CH_3$	ethanoic	ethanol	ethyl ethanoate
$CH_3CH_2COOCH_3$	propanoic	methanol	methyl propanoate
$HCOOCH_2CH_2CH_3$	methanoic	propan-1-ol	propyl methanoate
$CH_3COOCH_2CH_2CH(CH_3)_2$	ethanoic	3-methylbutan-1-ol	3-methylbutyl ethanoate

- Watch the order: CH_3COOCH_3 is methyl ethanoate, but $CH_3OOCCH_2CH_3$ (= $CH_3CH_2COOCH_3$) is methyl propanoate. Ethyl ethanoate and methyl propanoate are isomers ($C_4H_8O_2$); see the figure in 15.6.

MARK-SCHEME WORDING

- "3-methylbutyl ethanoate" (ALLOW "butanyl" for butyl; DO NOT AWARD "butanoyl").
- "2,2-dimethylpropyl propanoate" scores 2; "any name with '-propyl propanoate' scores 1".
- The alcohol named from an ester: non-systematic names are sometimes allowed if the structure is clear.

Often asked as: name an ester from its structure; name the alcohol and acid that make it; MCQ on ester names.

COMMON MISTAKE

- Writing the names the wrong way round (acid part first).
- Numbering the alkyl group from the wrong end: 3-methylbutyl, not 2-methylbutyl, when the branch is three carbons from the O.
- Calling $CH_3COO–$ "methanoate" (it's **ethanoate**: count the C=O carbon).

SPEC 15.14 Reactions of acyl chlorides

WHAT THIS SPEC POINT ASKS

Know the reactions of acyl chlorides with water, alcohols, concentrated ammonia and amines.

Why acyl chlorides are so reactive

- The carbonyl carbon is bonded to **two electronegative atoms (O and Cl)**, so it is strongly **δ+** and attacked readily by nucleophiles with a lone pair (O in water and alcohols, N in ammonia and amines).

- The overall change is **nucleophilic addition–elimination**: the nucleophile adds to the C=O, then **HCl is eliminated** (the Cl becomes Cl^-). In each case the –Cl is replaced.
- The reactions are fast and exothermic **at room temperature**, and give **steamy fumes of HCl** (with water and alcohols).

Reacts with	Equation (ethanoyl chloride)	Organic product
water	$CH_3COCl + H_2O \rightarrow CH_3COOH + HCl$	carboxylic acid
alcohol	$CH_3COCl + CH_3CH_2OH \rightarrow CH_3COOCH_2CH_3 + HCl$	ester
concentrated ammonia	$CH_3COCl + 2NH_3 \rightarrow CH_3CONH_2 + NH_4Cl$	primary amide (ethanamide)
primary amine	$CH_3COCl + 2CH_3NH_2 \rightarrow CH_3CONHCH_3 + CH_3NH_3^+Cl^-$	N-substituted amide (N-methylethanamide)

- With ammonia and amines, **two** molecules are needed: one reacts, and the second reacts with the HCl formed (giving NH_4Cl, a **white smoke/solid**, or an alkylammonium chloride).
- Making an ester from an acyl chloride is better than from the carboxylic acid: it is **fast**, needs no catalyst or heating, and goes to **completion** (not an equilibrium).

MARK-SCHEME WORDING

- 6-mark: equations for water → ethanoic acid + HCl, ethanol → ethyl ethanoate + HCl, ammonia → ethanamide + NH_4Cl, an amine → N-substituted amide; the **similarity**: all are addition–elimination (nucleophilic substitution) with HCl (or its salt) eliminated.
- Ammonia: "NH_3 reactant" and "NH_4Cl product and balanced equation", $CH_3COCl + 2NH_3 \rightarrow CH_3CONH_2 + NH_4Cl$.
- With concentrated ammonia: an amide forms, the reaction is exothermic, white smoke is seen.

Often asked as: equations for the four reactions; the product with a named nucleophile; the structure of an amide product; 6-mark comparing the reactions.

COMMON MISTAKE

- Writing HCl as the by-product with ammonia: it reacts with the excess NH_3, so write **NH_4Cl**.
- Giving NaOH when asked for the reagent that turns an acyl chloride into an acid (it's just **water**).
- Careless amide structures: two H on the N of an N-methyl amide, or two methyl groups.
- Not spotting the **similarities** between the reactions when a question asks for them.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Other nucleophiles** do the same thing: a thiol (CH_3SH) reacts like an alcohol to give a thioester (WCH14 Oct 2022).
- A **carboxylic acid** with an amine or ammonia gives an **ammonium salt**, not an amide (acid–base reaction) (WCH14 Jun 2024).

SPEC 15.15 Hydrolysis of esters

WHAT THIS SPEC POINT ASKS

Understand ester hydrolysis in acidic and in alkaline solution.

Acid hydrolysis

- **Heat under reflux** with **dilute** sulfuric or hydrochloric acid (the acid is a catalyst).
- It is the **reverse of esterification** and is **reversible**, so it doesn't go to completion:
  $CH_3COOCH_2CH_3 + H_2O \rightleftharpoons CH_3COOH + CH_3CH_2OH$
- Products: the **carboxylic acid** and the **alcohol**.

Alkaline hydrolysis

- **Heat under reflux** with **sodium hydroxide solution**:
  $CH_3COOCH_2CH_3 + NaOH \rightarrow CH_3COONa + CH_3CH_2OH$
- It goes **to completion** (one-way arrow). The carboxylic acid formed is immediately turned into the **carboxylate ion**, which can't react with the alcohol to re-form the ester.
- Products: the **sodium salt** of the acid and the **alcohol**. To get the free acid, add excess strong acid:
  $CH_3COONa + HCl \rightarrow CH_3COOH + NaCl$.

Splitting an ester

Break the bond between the C=O carbon and the single-bonded O. Add **OH** to the C=O side (acid) and **H** to the O side (alcohol). Methyl propanoate gives propanoic acid and methanol.

WORKED EXAMPLE

Worked example: purity of an ester by back titration

1.60 g of impure ethyl ethanoate (Mr 88.0) is refluxed with 50.0 cm^3 of 0.500 mol dm^{-3} NaOH (an excess). The leftover NaOH needs 16.0 cm^3 of 0.500 mol dm^{-3} HCl.

NaOH added = 0.500 × 0.0500 = 0.0250 mol; NaOH left = 0.500 × 0.0160 = 0.00800 mol

NaOH used by ester = 0.0170 mol = moles of ester (1 : 1)

mass of ester = 0.0170 × 88.0 = 1.50 g; purity = 1.50 ÷ 1.60 × 100 = **93.5%**.

MARK-SCHEME WORDING

- Triester (a fat or oil) with excess NaOH: "correct structure of propane-1,2,3-triol; correct structure of sodium ethanoate (or the salt); mole ratio" (3 NaOH, 3 salt). DO NOT AWARD a molecular formula.
- To release the acid: "add (excess) HCl(aq)" or any named strong acid.

Often asked as: products of acid and alkaline hydrolysis (MCQ); the reagent to free the acid from its salt; back-titration to find the percentage of ester; products of hydrolysing a triester.

COMMON MISTAKE

- Using ⇌ for alkaline hydrolysis, or → for acid hydrolysis.
- Giving the free acid (not the salt) as the product of alkaline hydrolysis.
- Sodium ethanoate drawn with a covalent O–Na bond.
- Answering "acid hydrolysis" when asked for a reagent: name the acid.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Back titration** of the excess NaOH to find the percentage of ester (WCH14 Jan 2020).
- **Triesters** (fats and oils) with excess alkali give propane-1,2,3-triol (glycerol) and three carboxylate salts (soaps) (WCH14 Oct 2022).

SPEC 15.16 Polyesters

WHAT THIS SPEC POINT ASKS

Understand how polyesters such as Terylene form by condensation polymerisation.

Condensation polymerisation

- In **condensation polymerisation**, monomers join and a **small molecule** (usually water, or HCl from an acyl chloride) is **lost** at each link.

- Each monomer needs **two** functional groups so the chain can keep growing at both ends.
- A **polyester** forms from:
  - a **dicarboxylic acid** (or diacyl chloride) + a **diol**, or
  - a single monomer with **–OH at one end and –COOH at the other** (a hydroxy acid, e.g. lactic acid).
- Each new link is an **ester link, –COO–**.

Terylene (PET)

- Monomers: **benzene-1,4-dicarboxylic acid**, HOOC–C_6H_4–COOH, and **ethane-1,2-diol**, $HOCH_2CH_2OH$.
- n HOOC–C_6H_4–COOH + n $HOCH_2CH_2OH$ → –[–OC–C_6H_4–COO–CH_2CH_2–O–]$_n$– + 2n H_2O
- Repeat unit: **–OC–C_6H_4–COO–CH_2–CH_2–O–** (one acid residue + one diol residue).

Poly(lactic acid)

- From 2-hydroxypropanoic acid, $CH_3CH(OH)COOH$: repeat unit **–O–CH(CH_3)–CO–**.
- Polyesters can be broken down by **hydrolysis** of the ester links (acid or alkali), so some (e.g. poly(lactic acid)) are **biodegradable**, unlike addition polymers such as poly(ethene).

Drawing and reading polymers

- Draw the repeat unit with **continuation bonds** at both ends (through the brackets) and **no –OH or –H** left on the ends.
- To find the monomers from a polymer: break each ester link; put **–OH** back on the C=O and **–H** on the O.

MARK-SCHEME WORDING

- "ester linkage"; "remainder of the structure including continuation bonds". ALLOW more than one repeat unit; IGNORE brackets and n.
- Poly(lactic acid) repeat unit "–[OCH(CH_3)CO]–" or "–[CH(CH_3)COO]–"; reaction type "esterification" (ALLOW condensation).
- Monomers "in either order"; OH groups written correctly on the monomers.

Often asked as: draw the repeat unit (or two repeat units) from given monomers; identify the monomers from a polymer section; MCQ on the type of polymerisation; hydrolysis of a polyester.

COMMON MISTAKE

- Ether, ketone or plain carbon links in place of the **ester** link.
- Leaving –OH or –COOH groups in the repeat unit, or extra OH groups branching off the chain.
- No continuation bonds; carbons with three or five bonds.
- Using a dialdehyde instead of a diacid as a monomer.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Polyesters from a **single hydroxy-acid** monomer (WCH14 Jan 2024), and the cyclic **lactide** formed from two lactic acid molecules (WCH14 Jan 2023). Terylene itself hasn't been named in a paper: questions give you the monomers.

SPEC 15.17 Mass spectra: accurate relative molecular mass

WHAT THIS SPEC POINT ASKS

Use mass spectra to suggest structures from accurate relative molecular masses, and calculate accurate Mr from relative atomic masses given to four decimal places.

High-resolution mass spectrometry

- The **molecular ion peak, M⁺**, is the peak with the highest m/z (ignoring small isotope peaks). Its m/z = Mr.
- A **high-resolution** spectrometer measures m/z to **4 decimal places**. Because accurate isotope masses aren't whole numbers (^{1}H = 1.0078, ^{16}O = 15.9949, ^{14}N = 14.0031, ^{12}C = 12.0000 exactly), molecules with the **same nominal Mr** have **different accurate Mr**.
- So the accurate Mr fixes the **molecular formula**. It can't tell **isomers** apart: they have identical formulae.

WORKED EXAMPLE

Worked example: which formula?

A compound has M^+ at m/z = 88.0524. Which is it: $C_4H_8O_2$, $C_5H_{12}O$ or $C_3H_8N_2O$?

$C_4H_8O_2$ = 4(12.0000) + 8(1.0078) + 2(15.9949) = **88.0522**

$C_5H_{12}O$ = 60.0000 + 12.0936 + 15.9949 = 88.0885

$C_3H_8N_2O$ = 36.0000 + 8.0624 + 28.0062 + 15.9949 = 88.0635

The closest match is **$C_4H_8O_2$** (e.g. ethyl ethanoate, methyl propanoate or butanoic acid). Other evidence (IR, NMR) is needed to choose between isomers.

Fragments tell isomers apart

- In the spectrometer, molecular ions break into **fragments**; only the **positive** fragments are detected.
- Propanal and propanone both have M^+ = 58, but **propanal** gives a peak at **m/z 29** ($C_2H_5^+$ or CHO^+), while **propanone** gives a big peak at **m/z 43** (CH_3CO^+) and none at 29.

Telling pentan-2-one from pentan-3-one

1. Both are $C_5H_{10}O$, so both give the molecular ion at m/z 86.
2. Pentan-3-one, $CH_3CH_2COCH_2CH_3$, breaks next to the C=O to give $[CH_3CH_2CO]^+$ at m/z 57 (and $C_2H_5^+$ at 29): $[CH_3CH_2COCH_2CH_3]^{+\bullet} \rightarrow CH_3CH_2CO^+ + \bullet CH_2CH_3$.
3. Pentan-2-one, $CH_3COCH_2CH_2CH_3$, gives a strong peak at m/z 43 from CH_3CO^+ and one at 71 from $CH_3CH_2CH_2CO^+$, but no peak at 57.

The same molecular ion but different fragments: fragmentation patterns tell isomers apart.

MARK-SCHEME WORDING

- Each fragment needs its m/z, its formula and a **positive charge**: "a fragment m/z = 136 (with a positive charge)". DO NOT AWARD a molecular formula or a negative charge.
- Fragmentation happens because "(high energy) electrons are fired at the molecules", "breaking bonds".
- "propanal and propanone have the same molecular ion peak m/z = 58"; "propanal has a peak at m/z 29 ($C_2H_5^+$ or CHO^+) but propanone does not".

Often asked as: MCQ choosing the compound from an accurate Mr; identify a fragment; use M^+ in a combined-spectra problem.

COMMON MISTAKE

- Saying two isomers have "the same mass spectrum": the M^+ is the same, but the **fragments** differ.
- Writing fragments with no + charge.
- Rounding accurate masses before comparing.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Distinguishing **propanal from propanone** using the fragment peaks at m/z 29 and 43 (WCH14 Jun 2025).

SPEC 15.18 Carbon-13 NMR: what it shows

WHAT THIS SPEC POINT ASKS

Understand that ^{13}C NMR gives information about the positions of ^{13}C atoms in a molecule.

How it works (in outline)

- About 1% of carbon atoms are **^{13}C**. Their nuclei behave like tiny magnets: in a strong magnetic field they absorb **radio waves** of a particular frequency (^{12}C nuclei don't).
- The exact frequency depends on the **electron environment** of each carbon. Electronegative atoms (O, N, Cl) nearby pull electrons away, so the carbon absorbs at a higher **chemical shift, δ**.
- δ is measured in **ppm** relative to the reference **TMS**, tetramethylsilane $Si(CH_3)_4$, set at δ = 0. TMS is used because it's inert, volatile (easily removed) and gives a single peak away from most others.
- Each **different carbon environment** gives **one peak**. Carbons in identical environments (related by symmetry) give the same peak.
- Samples are dissolved in a solvent with no interfering signals, such as $CDCl_3$.

MARK-SCHEME WORDING

- Answers must refer to **carbon environments**, not to hydrogen atoms.

Often asked as: as part of 15.19 questions; the idea isn't usually asked on its own.

COMMON MISTAKE

- Confusing ^{13}C and ^{1}H NMR, e.g. discussing splitting or peak areas for a ^{13}C spectrum.
- Mixing up IR ranges and NMR shifts.

SPEC 15.19 Using ^{13}C NMR spectra

WHAT THIS SPEC POINT ASKS

Use ^{13}C NMR data to predict the carbon environments in a molecule from chemical shifts, and justify the number of peaks.

Chemical shifts

Carbon environment	Typical δ / ppm
C–C (alkyl)	5–60
C–Cl, C–Br	30–70
C–N	35–60
C–O (alcohols, ethers, esters)	50–90
C=C, aromatic C	110–160
C≡N	110–125
C=O in acids and esters	160–185
C=O in aldehydes and ketones	190–220

(Use the ranges in your Data Booklet in the exam.)

Counting peaks

- **Number of peaks = number of different carbon environments.**
- Look for **symmetry**: carbons that can be swapped by a mirror plane or rotation are equivalent.

Compound	Environments (peaks)
propanone, CH_3COCH_3	2 (CH_3 × 2 equivalent; C=O)
propanal, CH_3CH_2CHO	3
butanone, $CH_3COCH_2CH_3$	4
pentan-3-one, $CH_3CH_2COCH_2CH_3$	3
2-methylpropanal, $(CH_3)_2CHCHO$	3
ethyl ethanoate, $CH_3COOCH_2CH_3$	4
2-methylpropan-2-ol, $(CH_3)_3COH$	2

WORKED EXAMPLE

Worked example: choosing an isomer

A ketone $C_5H_{10}O$ has **three** peaks in its ^{13}C spectrum, one at about 210 ppm. Which isomer is it?

Pentan-2-one, $CH_3COCH_2CH_2CH_3$, has 5 environments. 3-Methylbutan-2-one, $CH_3COCH(CH_3)_2$, has 4. **Pentan-3-one**, $CH_3CH_2COCH_2CH_3$, is symmetrical: CH_3 (×2), CH_2 (×2) and C=O, so **3 peaks**. The 210 ppm peak is the ketone C=O.

MARK-SCHEME WORDING

- "three peaks in its carbon-13 NMR spectrum; the other three have four peaks; labelling of the two equivalent carbon atoms". ACCEPT "three carbon environments".
- Shift answers need **both** the range and the environment: "190–225 ppm for C=O, 0–60 for C–C" (a value within the range is fine).

Often asked as: number of peaks for given structures; which isomer fits a given number of peaks; draw an isomer with a stated number of environments; assign a peak to a carbon.

COMMON MISTAKE

- Drawing a structure with more environments than the question allows (only about one in five got this right in one paper).
- Quoting a shift outside the allowed range, or with no environment.
- Not saying which carbon gives which peak.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Choosing between isomers from the **number of environments alone**, using symmetry (WCH14 Jan 2025).

SPEC 15.20 Proton (^{1}H) NMR

WHAT THIS SPEC POINT ASKS

Use low- and high-resolution proton NMR to predict proton environments from chemical shifts, relate peak areas to numbers of protons, deduce splitting with the n + 1 rule and suggest structures, and predict the spectrum of a given molecule.

What a ^{1}H spectrum tells you

Feature	Information
Number of peaks	number of different **proton environments**
Chemical shift, δ	the **type** of environment (what's nearby)
Relative peak areas (integration trace)	**ratio of the numbers of H** in each environment
Splitting (high resolution only)	number of H on the **adjacent** carbon(s)

- **Low resolution**: one peak per environment, no splitting. **High resolution**: peaks are split into multiplets.
- Areas give a **ratio**, not absolute numbers: 2 : 3 : 3 might be 2H : 3H : 3H.

Chemical shifts

Proton environment	Typical δ / ppm
alkyl H–C–C (CH_3, CH_2, CH)	0.8–1.8
H–C–C=O (e.g. CH_3CO–)	1.6–2.8
H–C–O (alcohol, ether, ester O–CH)	2.8–4.2
H–C–Cl, H–C–Br	3.0–4.2
O–H in alcohols	1–5 (variable)
H–C=C (alkene)	4.5–6.5
H on a benzene ring	6.5–8.0
aldehyde, –CHO	9–10
carboxylic acid, –COOH	10–12

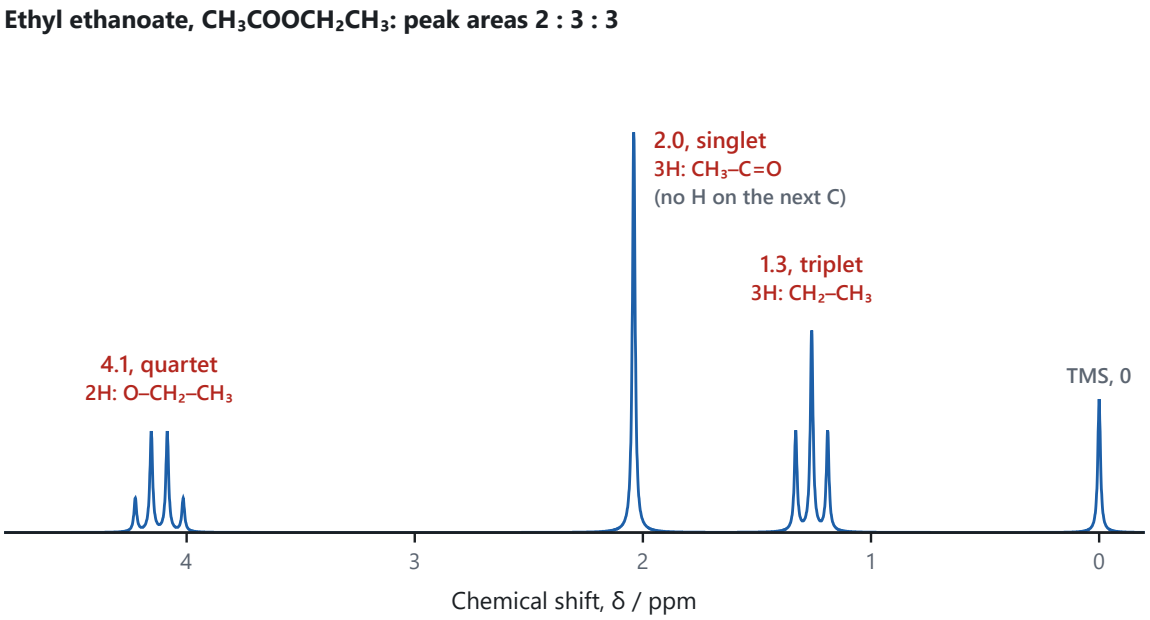
(Use the Data Booklet ranges in the exam.)

Splitting: the n + 1 rule

- A peak is split by the H atoms on the **neighbouring carbon(s)**. With **n** such H atoms, the peak splits into **n + 1** lines.
- H atoms in the **same** environment don't split each other. H on O (–OH) usually gives a **singlet** and doesn't split its neighbours.

H on adjacent carbons (n)	Lines	Name	Intensities
0	1	singlet	1
1	2	doublet	1 : 1
2	3	triplet	1 : 2 : 1
3	4	quartet	1 : 3 : 3 : 1
4	5	quintet (pentet)	1 : 4 : 6 : 4 : 1
6	7	heptet (septet)	

- Coupled groups come in pairs: an **ethyl group, CH_3CH_2–**, gives a **triplet (3H) + quartet (2H)**; an **isopropyl group, $(CH_3)_2CH$–**, gives a **doublet (6H) + heptet (1H)**.



High-resolution ^1H NMR of ethyl ethanoate (sketch). The CH_2 is split into a quartet by the CH_3 next to it; the CH_3 of the ethyl group is a triplet; the CH_3CO has no neighbouring H, so it's a singlet.

WORKED EXAMPLE

Worked example: ethyl ethanoate or methyl propanoate?

Both are $C_4H_8O_2$ and both give a singlet (3H), a quartet (2H) and a triplet (3H). The **shifts** decide:

- **Ethyl ethanoate, $CH_3COOCH_2CH_3$**: quartet at **4.1** (CH_2 joined to O, H–C–O), singlet at **2.0** (CH_3 next to C=O), triplet at 1.3.
- **Methyl propanoate, $CH_3CH_2COOCH_3$**: singlet at **3.7** (CH_3 joined to O), quartet at **2.3** (CH_2 next to C=O), triplet at 1.1.

So a quartet near 4.1 ppm and a singlet near 2.0 ppm (as in the figure) means ethyl ethanoate.

WORKED EXAMPLE

Worked example: propanal or propanone?

Propanal, CH_3CH_2CHO: 3 environments, areas **3 : 2 : 1**. CHO at 9–10 ppm (triplet, next to CH_2); CH_2 at about 2.4 ppm (split by 3 + 1 = 4 neighbours: a quintet/multiplet); CH_3 at about 1.1 ppm (triplet, next to CH_2).

Propanone, CH_3COCH_3: the two CH_3 groups are equivalent, so **one** peak (6H) at about 2.1 ppm, a **singlet** (no H on the C=O carbon next door).

WORKED EXAMPLE

Worked example: deducing a structure

C_3H_7Br gives two peaks: a doublet at 1.7 ppm and a heptet at 4.2 ppm, area ratio 6 : 1.

Two environments, ratio 6 : 1 → two equivalent CH_3 groups and one CH. The heptet means the CH has 6 neighbouring H; the doublet means each CH_3 has 1 neighbour. The CH at 4.2 ppm is next to Br.

Structure: **2-bromopropane, $CH_3CHBrCH_3$**. (1-Bromopropane would give three peaks, 3 : 2 : 2: a triplet, a sextet and a triplet.)

Sketching a low-resolution spectrum

One line per environment, heights in the **area ratio**, at sensible shifts, **no splitting**.

Identifying an unknown compound

1. A compound contains 54.5% carbon, 9.1% hydrogen and 36.4% oxygen by mass. Divide by A_r: C 54.5 ÷ 12 = 4.54, H 9.1 ÷ 1 = 9.1, O 36.4 ÷ 16 = 2.28. Divide by the smallest: C : H : O = 2 : 4 : 1, so the empirical formula is C_2H_4O (mass 44).
2. The mass spectrum has its molecular ion at m/z 88, so the molecular formula is $C_4H_8O_2$.
3. The IR spectrum has a strong absorption at 1740 cm^{-1} (C=O) but no broad absorption at 2500 to 3300 cm^{-1}, so there is no O–H: the compound is an ester, not a carboxylic acid.
4. The ^1H NMR spectrum shows three peaks: δ 4.1, quartet, area 2; δ 2.0, singlet, area 3; δ 1.3, triplet, area 3.

5 The quartet and triplet together are an ethyl group, and the CH_2 at δ 4.1 must be joined to oxygen (O–CH_2CH_3). The singlet of area 3 at δ 2.0 is a CH_3 next to C=O with no neighbouring hydrogens (CH_3CO–).

6 The compound is **ethyl ethanoate**, $CH_3COOCH_2CH_3$. Check: its ^{13}C NMR spectrum would show four peaks, including one near 171 ppm for the ester C=O.

MARK-SCHEME WORDING

- Explaining splitting must be **specific**: "4.2 ppm quartet due to 3 hydrogens on the adjacent carbon"; "1.3 ppm triplet due to 2 hydrogens on the adjacent carbon". General statements of the n + 1 rule don't score.
- Shift ranges quoted by mark schemes: "2.8–4.2 ppm (H–C–O)", "1.6–2.8 ppm (H–C–C=O)"; "CH_3CO … will be a singlet".
- "Peak areas show there are three proton environments in the ratio 3 : 2 : 1"; low-resolution sketch: correct number of peaks, heights in the ratio, sensible δ, **no splitting**.
- ALLOW pentet/quintet for 5 lines and hextet/sextet for 6.

Often asked as: assign each peak to a set of H; fill in a table of shift, splitting and relative area; explain a given spectrum; sketch a low-resolution spectrum; 6-mark deducing an isomer (often with mass spectrum and IR data too).

COMMON MISTAKE

- "Height" instead of "**area**"; "chemical environment" instead of "**proton** environment".
- Giving "3 protons in one environment and 1 in the other" instead of the **ratio**.
- Linking the shift to the functional group in general, not to the specific protons.
- Drawing split peaks for a **low-resolution** spectrum.
- Missing symmetry: quoting 3 : 2 : 1 for a symmetrical molecule.
- Treating the CH_3 next to C=O as the same environment as the CH_3 of an ethyl group.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Quintets** from 4 neighbouring H, e.g. the middle CH_2 of 4-hydroxybutanal (WCH14 Oct 2023). The n + 1 rule still applies; "quintet", "pentet" and "multiplet" are all accepted.
- Combined questions using M^+, IR, ^{13}C and ^1H data in one 6-mark answer (WCH14 Oct 2022). State which data confirm which feature.

SPEC 15.21 Chromatography: mobile and stationary phases

WHAT THIS SPEC POINT ASKS

Know that chromatography separates a mixture using a mobile phase and a stationary phase.

The principle

- The **stationary phase** stays in place: the solid silica or alumina layer on a TLC plate, the paper (and water held in its fibres) in paper chromatography, the packed solid in HPLC, the liquid coating in GC.
- The **mobile phase** moves through or over it, carrying the mixture: the solvent in paper chromatography, TLC and HPLC; an inert gas in GC.
- Each component is **shared (distributed) between the two phases**:
  - **stronger attraction to the stationary phase** (adsorption) → spends longer stuck → **moves less** (lower Rf, longer retention time);
  - **more soluble in the mobile phase** → **moves further** (higher Rf, shorter retention time).
- Different components have different balances of these, so they **separate**.

Polarity

- **Silica** (TLC stationary phase) is very **polar** (Si–OH groups). Polar molecules, especially ones that can hydrogen bond, are held strongly and move slowly.
- A **non-polar** solvent (e.g. hexane) moves non-polar compounds well, but leaves polar ones near the baseline; a more polar solvent moves polar compounds further.

MARK-SCHEME WORDING

- A compound moves a short distance because "only London forces form between it and hexane / it is polar and hexane is non-polar" and "weaker interaction between hexane and the compound / it is less soluble in hexane". DO NOT AWARD a mention of hydrogen bonds with hexane.
- Stationary phase with amine groups: the molecule with more OH groups is held more strongly because it "can form more / stronger hydrogen bonds (with the amine groups)".

Often asked as: MCQ on why one spot moves furthest; explain the position of a spot in terms of the two phases.

COMMON MISTAKE

- Discussing the solubility of the wrong pair (e.g. the solvent in the solvent) instead of the **compound in the mobile phase**.
- Just writing the Rf formula when the question asks you to **explain** a position.

SPEC 15.22 Rf values in paper and thin-layer chromatography

WHAT THIS SPEC POINT ASKS

Calculate Rf values from one-way paper and TLC chromatograms and understand why Rf values differ.

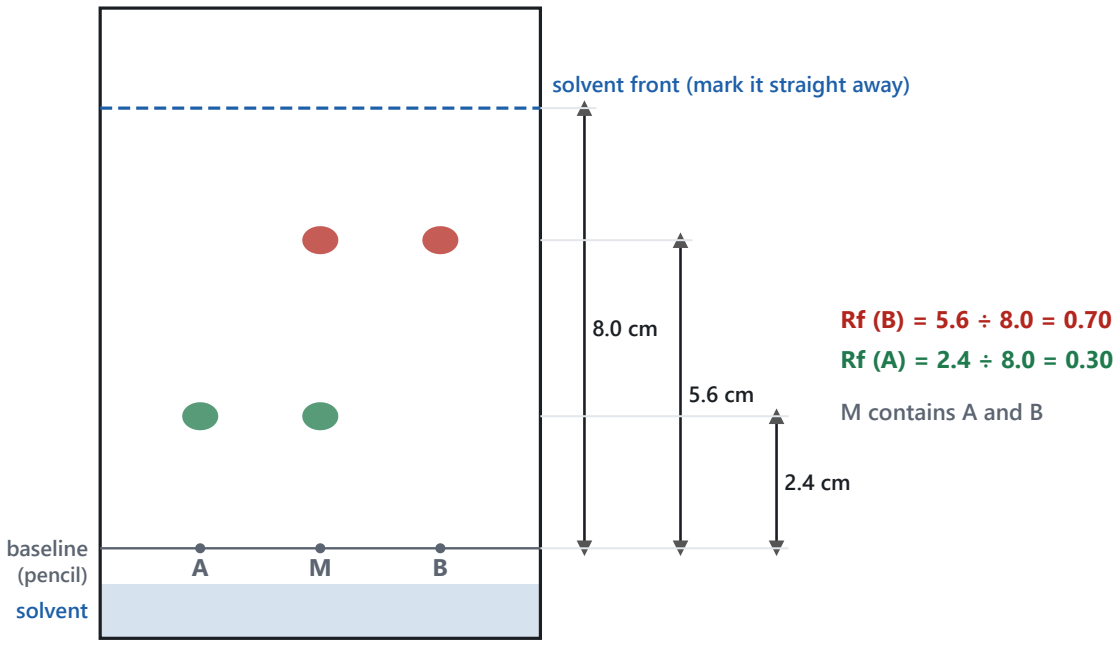
Running a chromatogram

1. Draw a **baseline in pencil** (ink would dissolve and run) about 1–2 cm from the bottom.
2. Put small, concentrated **spots** of the mixture and of pure reference compounds on the line.
3. Stand the plate/paper in a beaker with solvent **below the baseline** (so the spots don't dissolve into the solvent pool). **Cover** the beaker so the air is saturated with solvent vapour and the solvent doesn't evaporate off the plate.
4. Remove the plate when the solvent is near the top and **mark the solvent front** in pencil straight away (before it evaporates).
5. Dry the plate and **locate colourless spots**: UV light (TLC plates often have a fluorescent coating), iodine vapour, or ninhydrin spray for amino acids.

Calculating Rf

$$R_f = \text{distance moved by the spot} \div \text{distance moved by the solvent front}$$

- Measure **both distances from the baseline**, to the **centre** of the spot. Rf has **no units** and is always **between 0 and 1**.



A one-way TLC plate. Mixture M contains A and B, because its spots have the same Rf values as the reference spots run on the same plate.

WORKED EXAMPLE

Worked example: Rf

The solvent front is 8.0 cm above the baseline. Spot B has moved 5.6 cm.

Rf = 5.6 ÷ 8.0 = **0.70**.

Spot A (2.4 cm) has Rf = 0.30: it is held **more strongly by the stationary phase** (or is less soluble in the solvent) than B.

Why Rf values differ, and how to use them

- Rf depends on the compound's relative attraction to the stationary and mobile phases (polarity, hydrogen bonding, solubility).
- It also depends on **conditions**: the solvent, the type of plate or paper, and the temperature. So compare an unknown with **reference compounds run on the same plate** at the same time, rather than with a table.
- Two different compounds can have the same Rf in one solvent; running again in a different solvent helps.

MARK-SCHEME WORDING

- "Rf = distance moved by spot ÷ distance moved by solvent front" (distractors: dividing by the whole paper length; the reciprocal).

- Drawing a TLC plate for a given Rf: "origin / start line above the bottom of the paper"; "straight line for the solvent front, labelled"; spot at the right fraction of the distance. DO NOT AWARD a wavy line or the top of the paper as the front.

Often asked as: calculate Rf from a diagram or measurements (MCQ or 1–2 marks); draw a plate for a given Rf; why the baseline is in pencil; why the solvent must be below the baseline.

COMMON MISTAKE

- Measuring to the top of the paper instead of the solvent front, or from the bottom edge instead of the baseline.
- Freehand or wavy lines, no solvent front drawn, a spot impossibly close to the front.
- Not labelling which spot is which compound.

SPEC 15.23 HPLC and GC

WHAT THIS SPEC POINT ASKS

Know that HPLC and GC are column chromatography methods that separate substances by their different retention times, and that they can be linked to mass spectrometry (e.g. in forensics and drug testing in sport).

High-performance liquid chromatography (HPLC)

- **Stationary phase**: very small solid particles packed into a narrow steel **column** (small particles = large surface area = better separation).
- **Mobile phase**: a liquid solvent **pumped through at high pressure**.
- A **detector** (often UV absorbance) records each component as it leaves the column. The chromatogram plots **absorbance (y)** against **time (x)**.

Gas chromatography (GC)

- **Stationary phase**: a liquid with a high boiling temperature coated on a solid support, inside a long coiled column in an **oven**.
- **Mobile phase**: an **inert carrier gas** (helium, nitrogen or argon), which must not react with the sample.
- The sample is injected and **vaporised**. Components that are more volatile, or less attracted to the stationary liquid, come out first.

Reading a chromatogram

- **Retention time**: the time from injection to the component reaching the detector. Identify a component by comparing its retention time with a **pure reference sample** run under the **same conditions**.

- **Peak area** is proportional to the **amount** of that component.
- Retention time gets **longer** with a stronger attraction to the stationary phase, a slower flow of mobile phase, a longer column or (in GC) a lower temperature.

Linking to mass spectrometry

- Different compounds can have the **same retention time**, and some unknowns have no reference sample. So the column output is fed into a **mass spectrometer** (**GC–MS**, **HPLC–MS**).
- The chromatography **separates** the components; the mass spectrum (M^+ and fragmentation pattern) **identifies** each one, by comparison with a computer database.
- Uses: **forensics** (drugs, explosives, fire accelerants), **drug testing in sport** (urine and blood samples), food and environmental testing.

MARK-SCHEME WORDING

- "(use the pure compound to) find the retention time". IGNORE retention factor/Rf here.
- Name the technique for volatile liquids: "gas chromatography / GC / GLC". DO NOT AWARD several techniques.
- HPLC chromatogram axes: time on the x-axis, absorbance on the y-axis.
- Carrier gas: "inert".

Often asked as: what retention time means and how to use it; which technique suits a sample; how to reduce the retention time (MCQ: increase the carrier gas flow rate); why MS is added; scaling a sample mass for analysis.

COMMON MISTAKE

- Using **Rf** (paper/TLC) when the question is about HPLC or GC: they use **retention time** (only about one in five scored on this).
- Thinking GC or HPLC alone identifies an unknown with certainty.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Peak area** $\propto$ **amount** of each component, used to say which is most abundant (WCH14 Jun 2023).
- Preparing a standard solution for HPLC and scaling up the mass of a natural product (e.g. spinach for oxalic acid) needed for a given concentration (WCH14 Jan 2025).

Quick checks

QUICK CHECK

Which of these molecules has a chiral centre?

- propan-2-ol, $CH_3CH(OH)CH_3$
- butan-2-ol, $CH_3CH(OH)CH_2CH_3$
- propanone, CH_3COCH_3
- 2-methylpropan-2-ol, $(CH_3)_3COH$

In butan-2-ol, carbon 2 carries H, OH, CH_3 and C_2H_5: four different groups. In propan-2-ol the two CH_3 groups are the same.

QUICK CHECK

A single enantiomer of 2-bromobutane reacts with OH^- by an S_N2 mechanism. What is the product?

- a racemic mixture of butan-2-ol
- an optically inactive alcohol, because the carbocation is planar
- one enantiomer of butan-2-ol, with the arrangement inverted
- but-2-ene

S_N2 attack comes from the side opposite the bromine in one step, inverting the centre. There is no carbocation; a racemate would point to S_N1.

QUICK CHECK

Which reagent gives a positive result with propanal but not with propanone?

- 2,4-DNPH
- Tollens' reagent
- iodine in sodium hydroxide
- $LiAlH_4$

Only the aldehyde is oxidised, reducing Ag^+ to a silver mirror. 2,4-DNPH reacts with both; the iodoform test is positive for propanone (a methyl ketone) but not propanal.

QUICK CHECK

Why is the product of HCN addition to ethanal optically inactive?

- The product has no chiral centre
- HCN is a weak acid
- The reaction goes by S_N1
- CN^- attacks the planar C=O equally from both sides, forming a racemic mixture

2-hydroxypropanenitrile does have a chiral centre, but both enantiomers form in equal amounts, so their rotations cancel.

QUICK CHECK

Which reaction is the best way to make ethyl ethanoate quickly and completely?

- ethanoic acid + ethanol with a little concentrated H_2SO_4
- ethanoyl chloride + ethanol
- ethanoic acid + ethanal
- ethanoyl chloride + water

Acyl chlorides react with alcohols at room temperature, quickly and irreversibly. Acid-catalysed esterification with the acid is slow and reaches equilibrium.

QUICK CHECK

How many peaks are there in the ^{1}H NMR spectrum of methyl propanoate, $CH_3CH_2COOCH_3$, and what is the splitting of the CH_3 joined to oxygen?

- 2 peaks; triplet
- 3 peaks; quartet
- 3 peaks; singlet
- 4 peaks; singlet

There are three environments: CH_3 of the ethyl group (triplet), CH_2 (quartet) and O–CH_3. The O–CH_3 is a singlet because its neighbour across the oxygen is the C=O carbon, which carries no hydrogen.

QUICK CHECK

Which is NOT a reason for using TMS as the reference in NMR?

- it gives a single strong peak
- it is inert and has a low boiling point
- it dissolves the sample without giving any peaks
- its peak lies away from most other peaks

TMS is added in small amounts as a reference and gives a peak at δ 0. Dissolving the sample without adding peaks is the job of a solvent such as $CDCl_3$.

QUICK CHECK

On a TLC plate the solvent front moved 8.0 cm and a spot moved 2.6 cm. What is its R_f value?

- 3.1
- 0.26
- 0.33
- 5.4

$R_f = 2.6 \div 8.0 = 0.325 \approx 0.33$. R_f is always less than 1; 3.1 is the ratio upside down.